

PHISHING ATTACK

1.Scope:

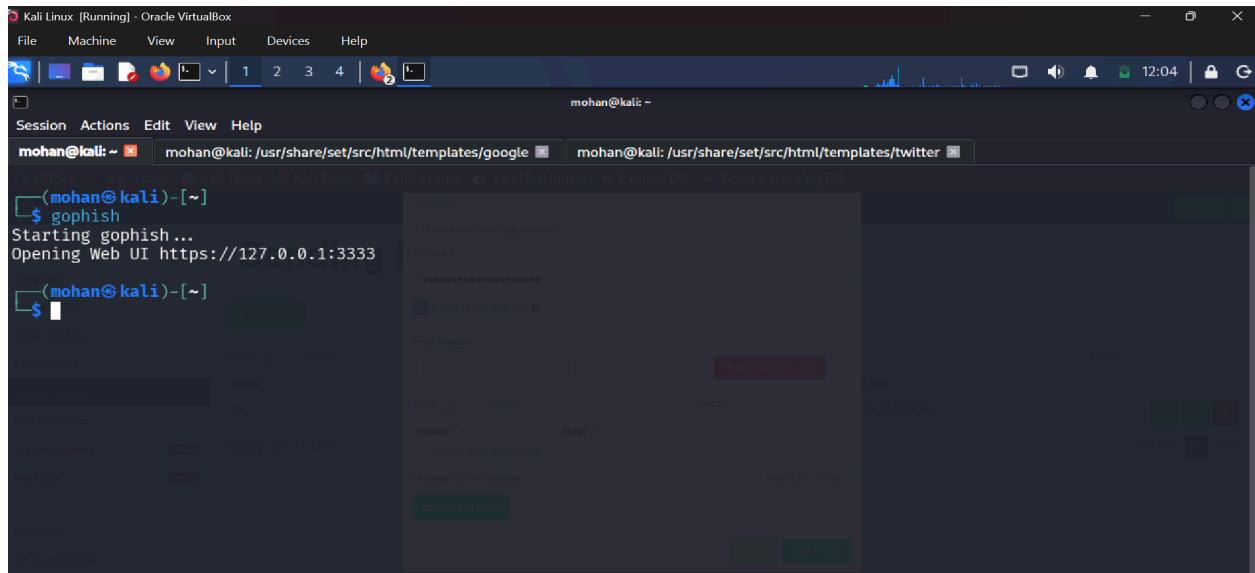
The scope of this project is to simulate a phishing attack in a controlled lab environment using **Kali Linux** and **GoPhish** with a **Gmail** account for sending emails. The purpose of the simulation is to understand how phishing attacks are carried out and how attackers trick users into revealing sensitive information such as usernames and passwords. This project also aims to demonstrate how phishing attacks can be detected and prevented. The experiment is conducted in a **controlled environment for educational purposes only** and does not target real users.

2.Methodology:

The methodology followed in this project includes setting up the phishing framework, configuring email settings, designing phishing pages, creating user groups, and launching a phishing campaign.

2.1 Environment Setup

1. Install **Kali Linux** on a virtual machine or system.
2. Download and install **GoPhish** phishing framework.
3. Start the GoPhish server and access the admin dashboard through a web browser.
4. Log in using the default credentials and configure the dashboard.



2.2 Sending Profiles

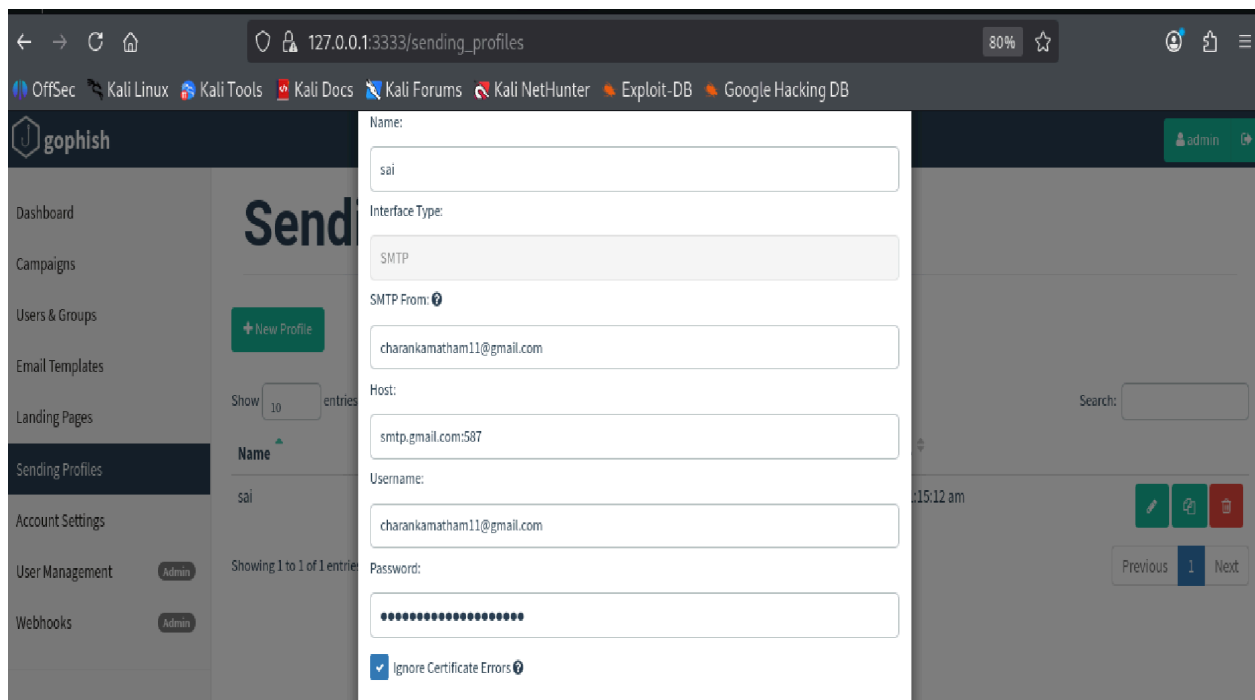
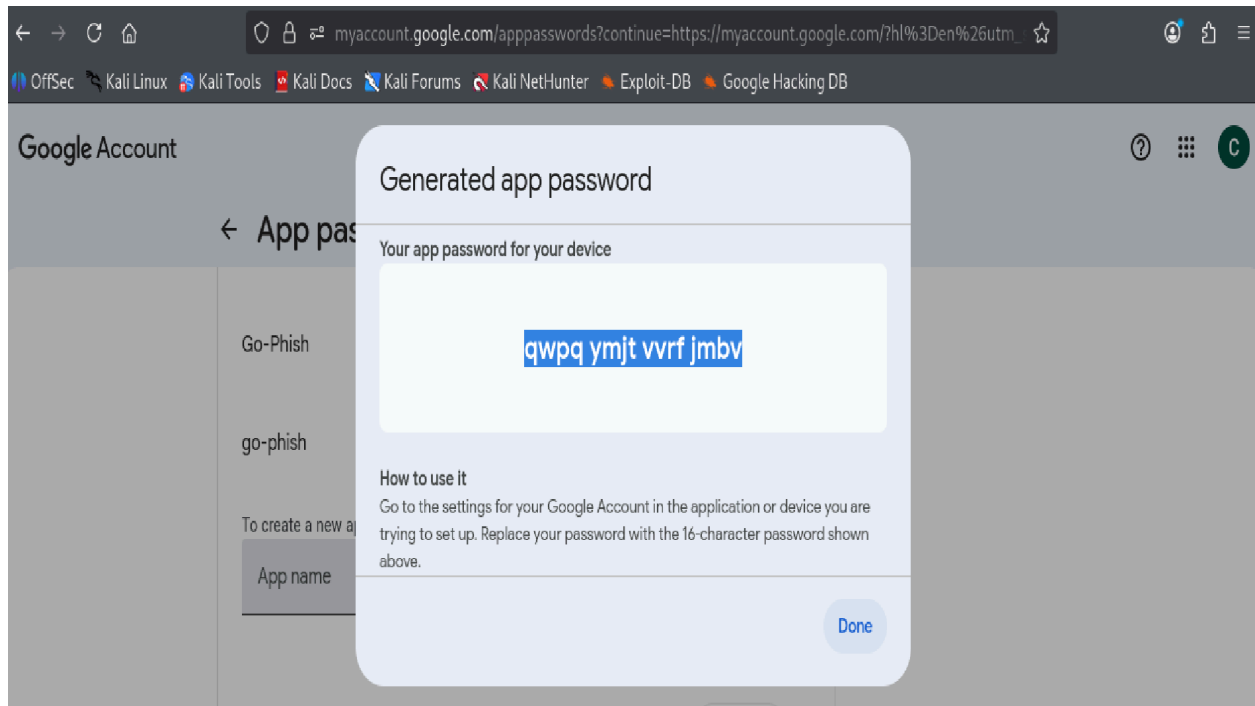
Sending profiles are used to configure the email service that will send phishing emails.

Steps followed:

1. Navigate to **Sending Profiles** in the **GoPhish** dashboard.
2. Create a new sending profile.
3. Configure SMTP settings using **Gmail**.

Configuration includes:

- Email address
 - SMTP server
 - Port number
 - Username and password
 - TLS settings
4. Test the email configuration to verify that the system can successfully send emails.

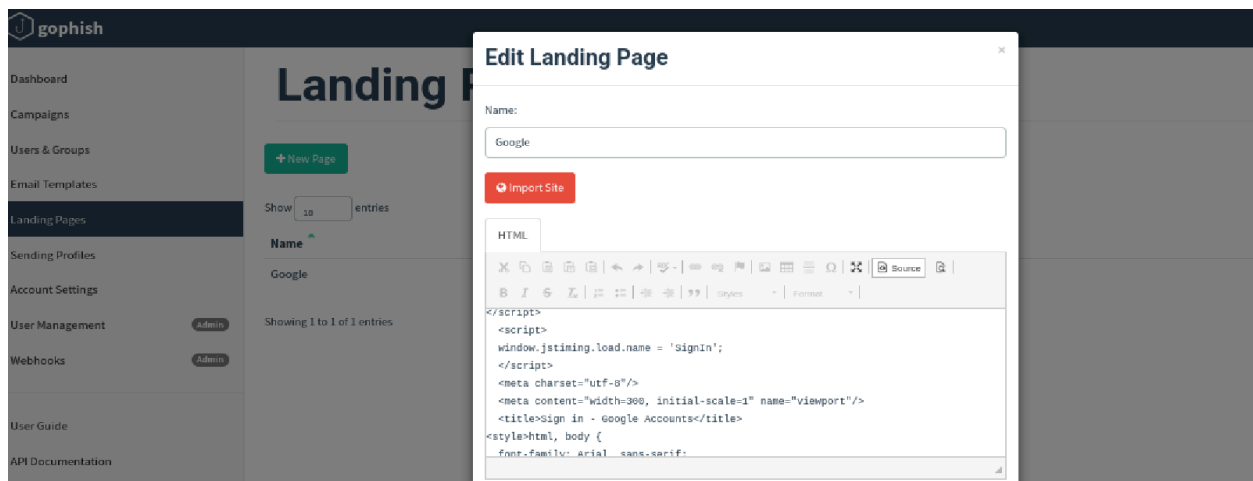


2.3 Landing Pages

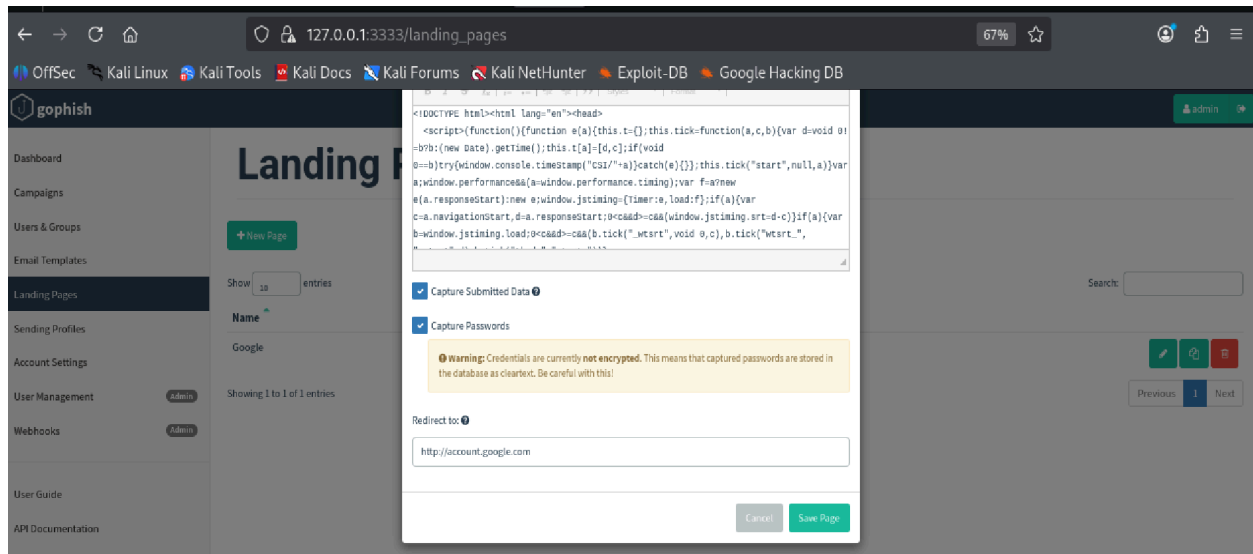
Landing pages are fake websites that capture user credentials.

Steps followed:

1. Navigate to **Landing Pages** in **GoPhish**.
2. Create a new landing page template.
3. Design a login page similar to a legitimate service login page.
4. Enable **Capture Submitted Data** to record credentials entered by users.
5. Save the landing page configuration.



```
mohan@kali: /usr/share/set/src/html/templates/google
Session Actions Edit View Help
mohan@kali: ~
mohan@kali: /usr/share/set/src/html/templates/google
mohan@kali: /usr/share/set/src/html/templates/twitter
$ cd ..
(mohan@kali)-[/usr/share/set/src/html/templates]
$ ls
google __init__.py java __pycache__ template.py twitter
(mohan@kali)-[/usr/share/set/src/html/templates]
$ cd google
(mohan@kali)-[/usr/share/set/src/html/templates/google]
$ ls
index.template
(mohan@kali)-[/usr/share/set/src/html/templates/google]
$ open index.template
(mohan@kali)-[/usr/share/set/src/html/templates/google]
$ open index.template
(mohan@kali)-[/usr/share/set/src/html/templates/google]
$
```

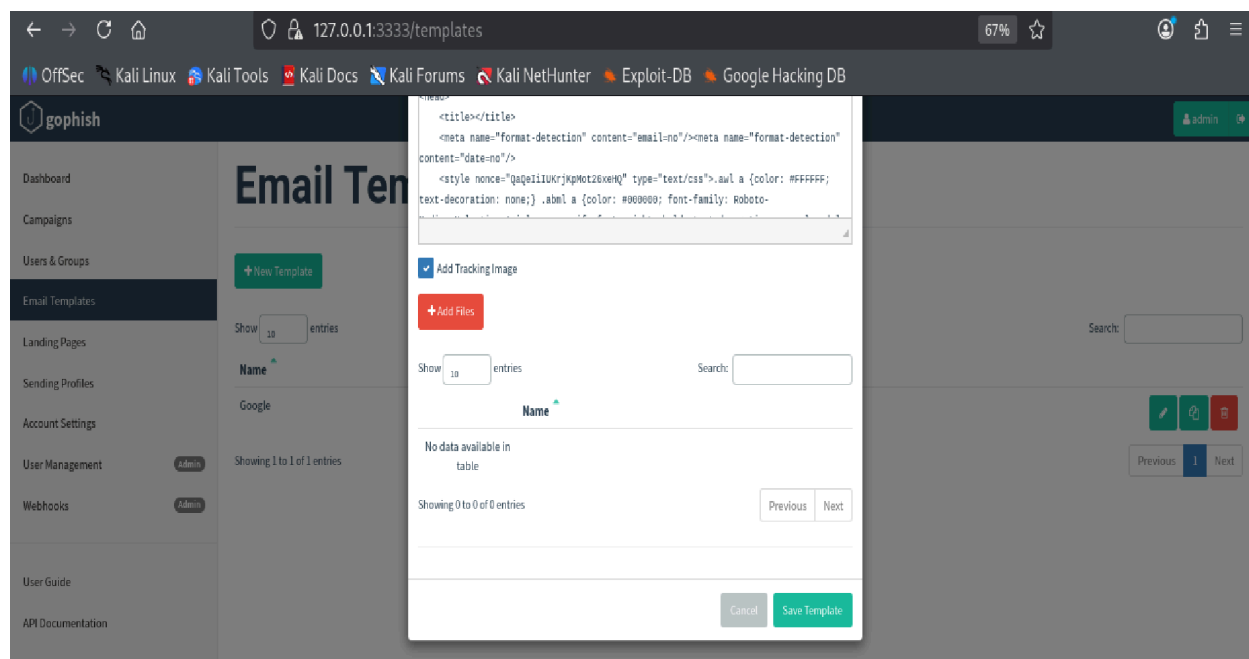
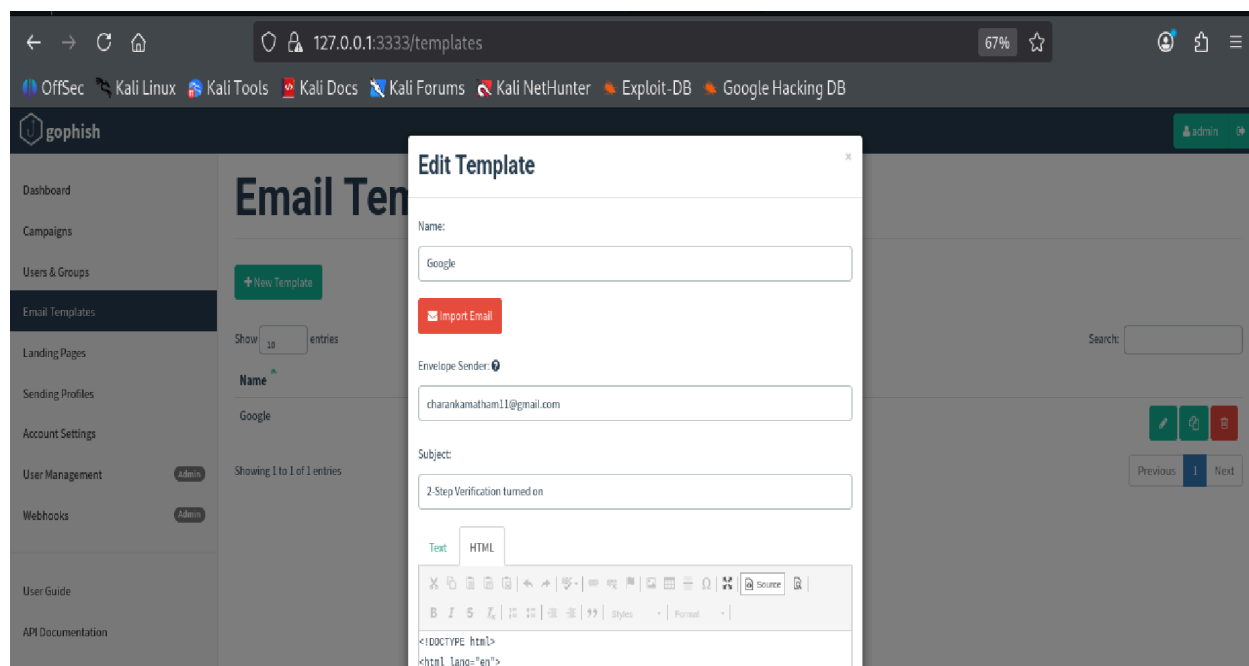


2.4 Email Templates

Email templates are designed to imitate legitimate emails to attract users to click phishing links.

Steps followed:

1. Navigate to **Email Templates** in the **GoPhish** dashboard.
2. Create a new email template.
3. Design the email content to resemble a legitimate notification such as:
 - Account verification
 - Password reset request
 - Security alert
4. Insert phishing link pointing to the fake login page.

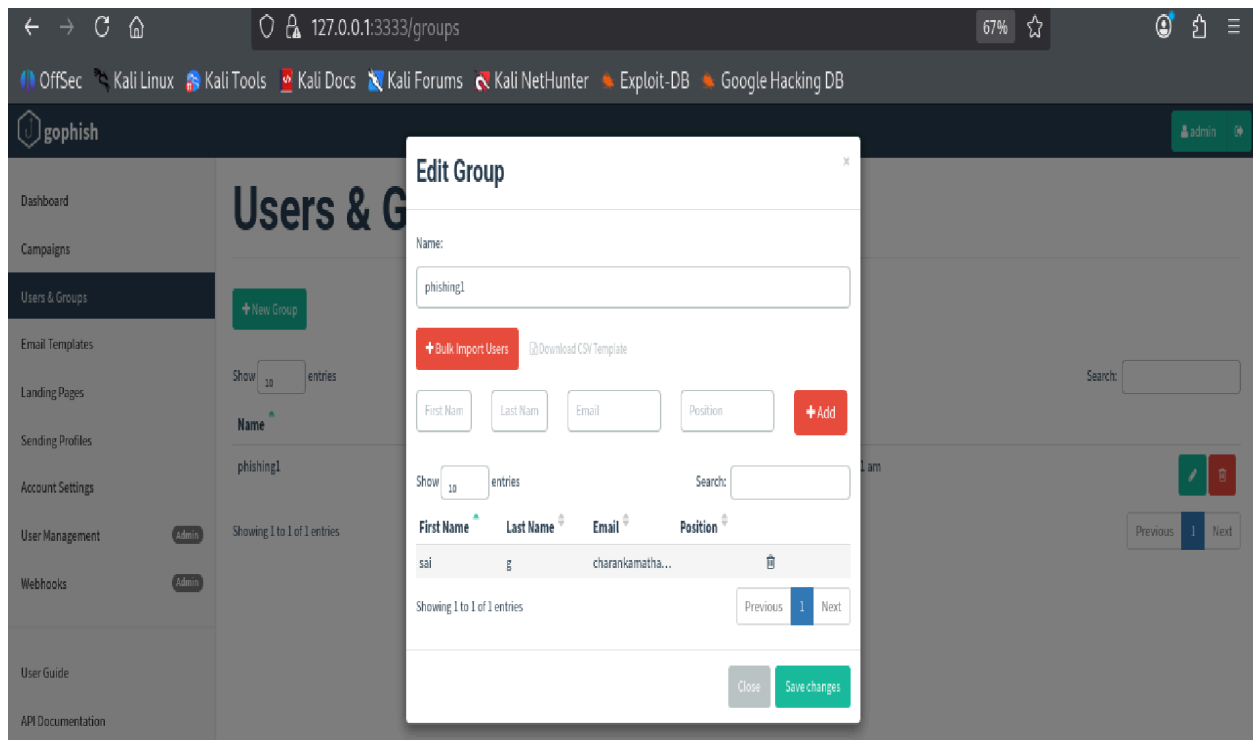


2.5 Users and Groups

Users and groups represent the target audience for the phishing campaign.

Steps followed:

1. Navigate to **Users & Groups** in **GoPhish**.
2. Create a new group for testing purposes.
3. Add test users with the following details:
 - Name
 - Email address
 - Position
4. Save the group to use in the phishing campaign.



2.6 Campaign Creation

The campaign integrates all configured components.

Steps followed:

1. Navigate to **Campaigns** in **GoPhish**.
2. Create a new campaign.
3. Configure campaign settings including:
 - Campaign name
 - Sending profile
 - Email template
 - Landing page
 - Target user group
4. Launch the phishing campaign.

The screenshot displays the GoPhish web application interface. On the left is a sidebar menu with options: Dashboard, Campaigns (selected), Users & Groups, Email Templates, Landing Pages, Sending Profiles, Account Settings, User Management, Webhooks, User Guide, and API Documentation. The main content area is titled 'Campaigns' and features a '+ New Campaign' button. Below this, there are tabs for 'Active Campaigns' and 'Archived Campaigns'. A table shows one active campaign named 'Go-phishing'. A modal form for creating a new campaign is open in the center. The form fields are: Name (Go-Phishing), Email Template (Google), Landing Page (Google), URL (http://10.1.171.157), Launch Date (March 13th 2026, 12:18 pm), Send Emails By (Optional), Sending Profile (sai), and Groups (phishing2). At the bottom of the modal are 'Close' and 'Launch Campaign' buttons. The browser's address bar shows '127.0.0.1:3333/campaigns'.

127.0.0.1:3333/campaigns

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

gophish

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management

Webhooks

User Guide

API Documentation

+ New Campaign

Active Campaigns

Archived Campaigns

Show 18 entries

Name

Go-phishing

Showing 1 to 1 of 1 entries

Name

Go-Phishing

Email Template

Google

Landing Page

Google

URL

http://10.1.171.157

Launch Date

March 13th 2026, 12:18 pm

Send Emails By (Optional)

Sending Profile

sai

Groups

phishing2

Close

Launch Campaign

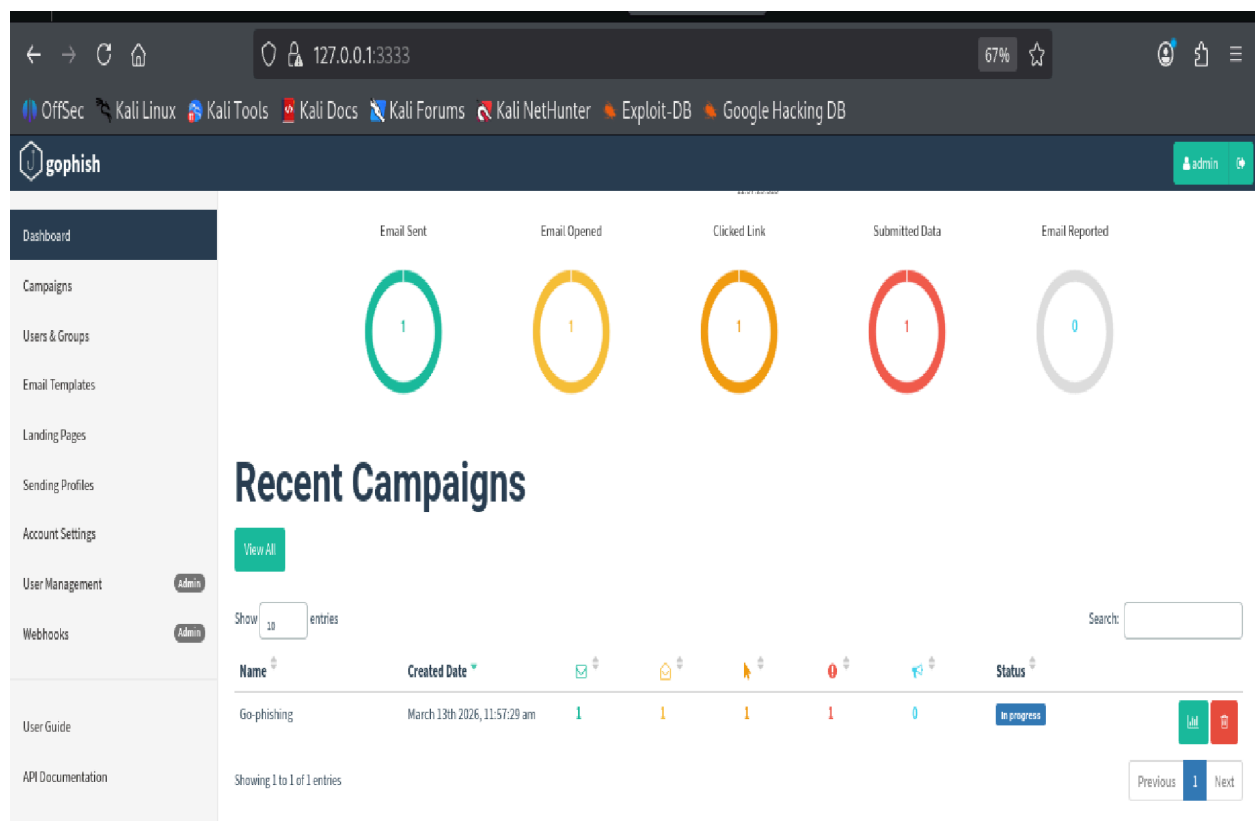
2.7 Monitoring Campaign Results

After launching the campaign, results are monitored through the dashboard.

The system tracks:

- Emails sent
- Emails opened
- Link clicks
- Credentials submitted

The analytics provided by **GoPhish** help analyze how users interact with phishing emails.

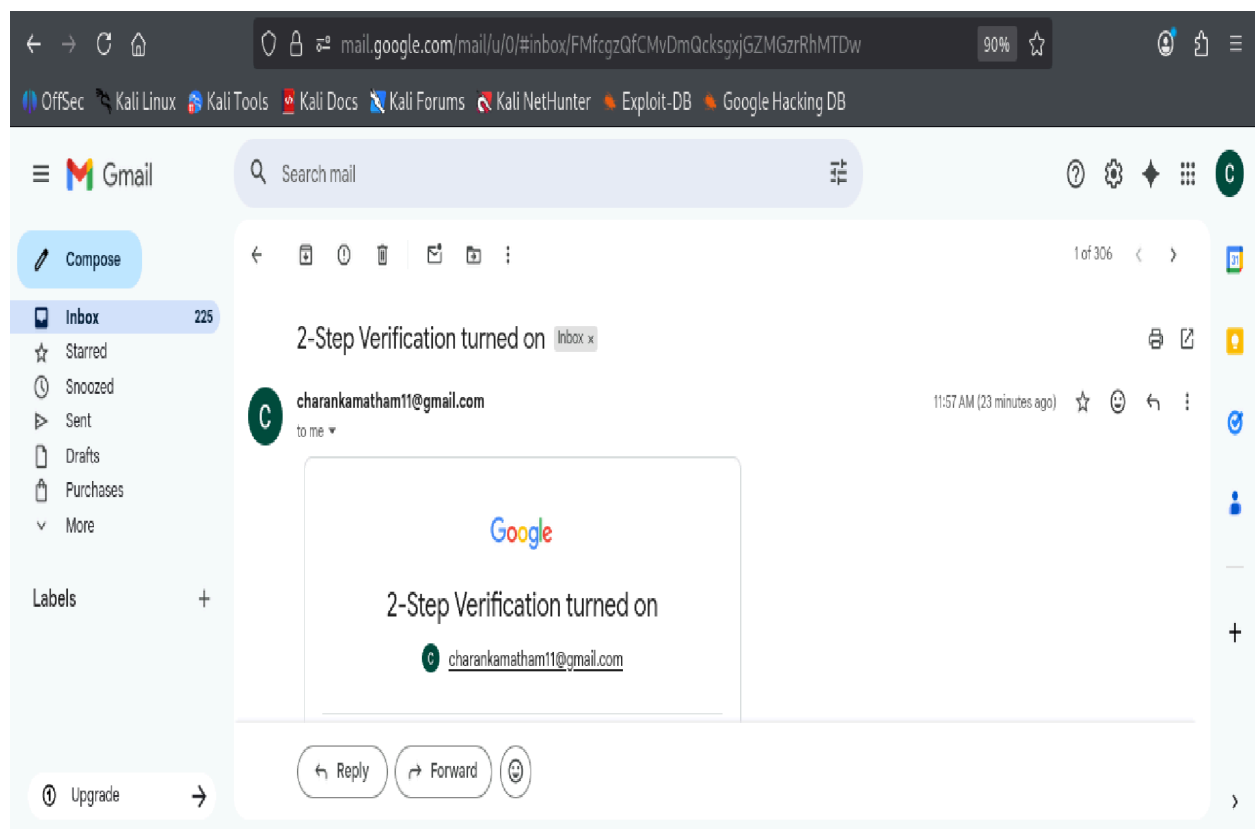


3. Findings

The phishing simulation produced the following observations:

- Phishing emails were successfully delivered using **Gmail** SMTP configuration.
- Some users opened the phishing email and clicked the link.
- The fake login page successfully captured submitted credentials.
- The campaign dashboard displayed detailed statistics of user interactions.

These results demonstrate how phishing attacks can successfully trick users into revealing sensitive information.





Sign in with your Google Account



Sign in
[Need help?](#)

 admin

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management Admin

Webhooks Admin

User Guide

API Documentation

Details

Show entries

Search:

First Name	Last Name	Email	Position	Status	Reported
sai	g	charankamatham11@gmail.com		Submitted Data	

Timeline for sai g

Email: charankamatham11@gmail.com
Result ID: JgvPRpx

 Campaign Created

March 13th 2026 11:57:29 am

 Email Sent

March 13th 2026 11:57:32 am

 Clicked Link

March 13th 2026 11:57:44 am

 Linux (OS Version: x86_64)

 Firefox (Version: 140.0)

The screenshot shows the gophish web interface in a browser. The address bar displays '127.0.0.1:3333/campaigns/4'. The interface includes a sidebar with navigation options: Dashboard, Campaigns, Users & Groups, Email Templates, Landing Pages, Sending Profiles, Account Settings, User Management (with an 'Admin' button), Webhooks (with an 'Admin' button), User Guide, and API Documentation. The main content area shows 'Submitted Data' for a campaign on 'March 12th 2026 11:58:15 am'. It lists the operating system as 'Linux (OS Version: x86_64)' and the browser as 'Firefox (Version: 140.0)'. A green 'Replay Credentials' button is visible. Below this, a 'View Details' section contains a table of parameters and their values.

Parameter	Value(s)
Email	sai1234@gmail.com
GHLX	SJLChgagoM
PersistentCookie	yes
_urlB	0
bgresponse	js_disabled
checkConnection	
checkedDomains	youtube
continue	https://accounts.google.com/o/oauth2/auth?zt=ChRsWFBwd2JmV1hlcDHUFDic2BENhIFVWsaStdNLW9MdThibW1TMFQzVUZfc1BBaURuNmIRSQhE2%88%69APsEzfgAAAAUy4_qD7Hbfz38w8konalNoulcRiD3YTX
dnConn	
dsh	-7381887106725702428
password	sai@1234

4. Fixing / Prevention Measures

To prevent phishing attacks, organizations should implement the following measures:

Security Awareness Training

Employees should be trained to recognize suspicious emails and avoid clicking unknown links.

Email Filtering

Enable spam filtering and phishing detection mechanisms in **Gmail** or enterprise email systems.

Multi-Factor Authentication

Enable MFA for all accounts to reduce the risk of unauthorized access.

Email Authentication Protocols

Implement:

- SPF (Sender Policy Framework)
- DKIM (DomainKeys Identified Mail)
- DMARC (Domain-based Message Authentication)

Security Monitoring

Regular monitoring and phishing simulations should be conducted using tools like **GoPhish**.

5. Conclusion

The phishing simulation conducted using **Kali Linux** and **GoPhish** demonstrated how phishing campaigns are created and executed. The experiment highlights the vulnerability of users to deceptive emails and fake login pages.

The results emphasize the importance of **cybersecurity awareness, proper email security mechanisms, and multi-factor authentication** in preventing phishing attacks.

Organizations should implement regular training and technical safeguards to reduce the risk of successful phishing attempts.